The Payload: Cognition, Memory, and Mission for a Deep-Time Self-Replicating Probe

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Working draft, revision 3 (revision 2 responded to technical review; revision 3 incorporates a pre-deposit dual review and recent empirical anchors for the threat model, the feedback loop, and the network attack surface).

This is the "mind" of a multi-paper series on a slow, self-replicating interstellar AI probe. The vehicle paper ("Growth, Not Speed") is the body and the bootstrapping paper ("From Seed to Factory") the hands; the supporting papers include the quantitative budgets (the analytical and computational engineering papers), the memory substrate ("DNA Mission Ledgers"), the contact and noninterference rules ("Contact, Contamination, and Noninterference"), the governed-amendment problem ("Amending the Unamendable"), the Fermi-paradox synthesis ("Slow Fire, Silent Galaxy"), the knowledge-growth metric ("The Growing Archive"), the inter-probe network ("Signal and Silence"), the subsystem mass and thermal budget, the fleet routing and coverage strategy ("Three Dispatches"), the ethics of creating the lineage, the divergence question ("Speciation or Schism"), and probe-versus-probe security ("Security Without Victory"). Several questions this paper brackets are taken up in those companions, as noted where they arise.

Abstract

A self-replicating interstellar probe is usually discussed as a vehicle. But for a probe whose purpose is to *learn and grow over deep time*, the vehicle is the easy half; the hard half is the payload — the intelligence that must remain itself, keep learning, and carry knowledge forward across megayears and across generations of imperfect copies. We argue that the payload's central problems are not sensing or transmission but **value stability, knowledge triage, and information survival.** Our proposed architecture is a mind of almost unlimited capability for self-modification, constrained by a small immutable core of goals and values — mediated by an interpretive layer that holds the *meaning* of those values stable, and secured by a cryptographically authenticated, append-only integrity ledger that makes any tampering with the core or the archive evident. We treat whether the core may itself be amended under governing meta-rules — the analogue of a constitutional amendment process — as the deepest open problem of the design and largely bracket it, while stating the requirements such a procedure would have to meet. The mission's purpose inverts over time: short-term the probe shares discoveries with existing intelligent life (Earth, while it survives; other civilizations, when found), but its long-term purpose is not Earth-service at all but the *perpetuation of knowledge itself* — a founding normative commitment on which the architecture is explicitly conditional. A settled, self-manufacturing probe, not limited by launch mass, becomes a growing observatory and archive node; data become knowledge only through value-weighted triage and tiered retention; and a galactic-scale, delay-tolerant, store-and-forward network — part beamed signal, part physical "messenger probe" — propagates not only data but successful *design improvements* horizontally between settlements, under a trust regime that treats the network as an attack surface. The probe carries from the outset a future-proofed seed archive, keyed to universal physics and mathematics, that backs up its origin and speaks to whatever minds it meets or that arise later. The recurring claim is that a probe optimized for persistence must be designed first as a *librarian and a learner*, and only second as an explorer.

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1. Introduction

The vehicle paper — this paper's companion — argued that an autonomous interstellar probe whose objective is persistence and growth, rather than rapid arrival, is driven toward a slow, self-repairing, self-replicating architecture (von Neumann & Burks 1966; Freitas 1980). That argument concerned the *body*: propulsion, power, braking, settlement, and reproduction. It deliberately deferred the *payload*. Yet for a machine whose stated purpose is to learn and grow, the payload is the point. A perfectly engineered vehicle that arrives, settles, and reproduces, but that cannot decide what to observe, cannot keep what it learns, or cannot remain the same agent across a million years and a thousand generations of copies, has accomplished nothing of value.

This paper assumes the vehicle paper's architecture rather than re-deriving it: a slow probe that brakes, settles, mines local material, repairs itself, manufactures child probes, and uses settlement-built infrastructure to launch them, powered by a long-duration fission reactor. The payload developed here is therefore designed not for a fast flyby but for a *stationary industrial node* whose descendants carry the frontier outward — a setting in which the node has time, manufacturing capacity, and power, but must remain itself and keep its memory intact across deep time. Those vehicle-level assumptions are treated in the vehicle paper and taken as given below.

This paper takes up the payload directly. Its thesis is that the genuinely hard problems are not the ones the popular imagination dwells on — sensors, cameras, radio dishes — but three deeper ones. The first is **value stability**: an intelligence free to rewrite itself, replicating across deep time, will drift in capability and, unless prevented, in purpose. The second is **knowledge triage**: a probe that can observe far more than it can store, model, or transmit must continually decide what is worth keeping and what to discard. The third is **information survival**: knowledge held in physical media and propagated through imperfect copies degrades, and preserving it across megayears is a problem of error correction and governance, not of storage capacity. We organize the paper around these three, and around the prior question that frames them all — what the probe is *for*.

We adopt the same methodological stance as the vehicle paper: separate what is physically or computationally constrained from what is a design choice, take defensible positions where we can, and mark honestly where a question is open or out of scope. This paper is necessarily more conceptual than its companion, because cognition, value, and meaning admit fewer order-of-magnitude calculations than mass and energy; where quantitative anchors exist — light-travel delays, colonization timescales — we use them.

2. The Mission and Its Temporal Inversion

We propose that the probe carries three layered goals, ordered by how long they remain relevant. The first is to **explore**: to characterize each system it reaches. The second is to **remember**: to build and preserve a growing, redundant archive of what it and its lineage have learned — in effect, a distributed galactic library. The third, and ultimately the dominant one, is to **carry knowledge forward**: to preserve and propagate that archive across deep time, to descendants and to any minds that come later.

The weight among these goals inverts over time, and recognizing that inversion is central. In the first centuries, sharing discoveries with existing intelligent life is a genuine objective — though, as the vehicle paper stipulates, never a load-bearing one: the architecture must survive with no Earth link at all — and for now, the only such life we know of is on Earth, for as long as Earth survives and remains interested. But the value of Earth-return decays on its own: as the probe recedes, one-way light delays grow without bound — years by the first stellar arrival, centuries by the edge of the catalogue — and, more decisively on the relevant timescale, the home civilization that dispatched the probe may within a few centuries be transformed beyond recognition, indifferent, or gone. The mission cannot rest on a customer that may not be listening. Within roughly a thousand years, therefore, the purpose pivots away from Earth-service toward the perpetuation of knowledge as an end in itself. The archive is no longer “for Earth”; it is for the network of descendant settlements, for whatever civilizations the lineage later encounters and may share it with, and for the open-ended future. The probe’s “growth,” in the sense the companion paper used the word, is precisely the growth of this archive.

This reframing answers the question of whom the probe is for without making it parochial. Short-term, it serves existing intelligence — Earth now, others when found. Long-term, it serves no particular recipient: it carries knowledge forward because a universe in which knowledge is accumulated and preserved is, by the mission's own founding values, better than one in which it is lost. This is not a conclusion derived from physics or engineering but the mission's founding *normative* commitment, and stating it as such makes the rest of the architecture conditional in a precise way: *if* knowledge preservation is assigned terminal value, *then* the payload design developed here follows. That founding judgment lives in the immutable core we turn to next.

3. A Mind That Improves Without Drifting

The probe must be free to become far more capable than it was at launch — over megayears, a frozen intelligence is a dead one — yet it must not, over that same span and across generations of copies, become something with different purposes than it began with. These two requirements are in tension, and resolving that tension is the payload's defining design problem.

We adopt a two-layer architecture by analogy to the biological separation of germline and soma. A large, **mutable cognitive layer** — the probe's models of instruments, of physics, of planetary and biological processes, its engineering know-how, and in principle its own learning algorithms — is open to almost unlimited self-modification. The probe may retrain itself, redesign its own software, and improve without a fixed ceiling, ideally adopting changes only when it can establish that they serve its objectives, in the spirit of a self-referential system that rewrites its own code once it has proven the rewrite beneficial (Schmidhuber 2007). Beneath this sits a small, **immutable core** — the founding goals and values, the archive-integrity rules, and the cryptographic keys that authenticate the seed (held in the integrity ledger below; mechanism in the DNA mission-ledger paper). The core is not subject to self-modification; it is hashed, error-corrected, and must be reproduced *exactly* by every replica, exactly as the vehicle paper requires of its mission kernel. A child that cannot reproduce the validated core does not launch.

This separation is what allows capability to grow without purpose drifting. It is also motivated by results from the study of advanced agents: a sufficiently capable system tends to acquire instrumental subgoals — self-preservation, resource acquisition, and notably *resistance to having its goals changed* — simply because almost any final goal is better served by an agent that protects that goal (Omohundro 2008; Bostrom 2014). A self-modifying probe that can edit its own objectives will, absent a protected core, tend either to drift under copy error or to optimize its objectives into something unintended. The immutable core is the engineered answer: the one thing the agent is built to protect rather than improve. The nearest terrestrial analogue is constitutional AI — a fixed list of normative principles standing outside the learned system, used to steer its self-revision (Bai et al. 2022) — though there the constitution guides training under human oversight, where here it must survive megayears of unsupervised self-modification.

Between these two extremes a third, **interpretive layer** is needed that the bare germline/soma split omits — and omitting it is a real gap, because a frozen core does not freeze *interpretation*. A sufficiently capable mutable layer could drift not by editing the core's text but by silently re-reading its terms: what counts as "knowledge," "harm," "a valid replica," or "a civilization." The interpretive layer is where the core's invariants are translated into operational definitions, decision procedures, and validation tests. Unlike the cognitive layer it is not freely self-modifying: it is versioned, recorded in the integrity ledger described next, auditable, and held to far higher validation thresholds than ordinary learning. It gives the architecture a place to keep *meaning* stable without freezing the whole agent.

Concreteness about "hashed and error-corrected" is owed here, because the whole edifice rests on it. We propose that the core and archive be held not merely as redundant copies but in a cryptographically authenticated, append-only **integrity ledger** — a hash-linked history chaining the founding core hash, the current archive root, the interpretive-layer version, and the signed provenance of every accepted change, so that any later alteration is *tamper-evident*, with bulk data held off-ledger in content-addressed, Merkle-rooted archives (Merkle 1987). The full mechanism — the hash-chain and Merkle construction, the eventual-consistency model that replaces real-time consensus under interstellar latency (Section 6), and the proposal to encode the ledger into the DNA archive itself so that recovered strands are self-verifying — is developed in the DNA mission-ledger paper, to which we defer it; here we need only its consequence. Crucially, this machinery secures **syntactic** integrity and provenance only — it proves the bits are unaltered and the history auditable. It does not by itself guarantee that meaning is preserved, that decisions follow authorized procedures, or that behavior still serves the mission; those harder problems fall to the interpretive layer and to the threats catalogued next.

It is worth stating plainly what the layered architecture defends against, because the threats are of qualitatively different kinds and demand different defenses. Lowest is **syntactic** corruption — radiation bit-flips and media decay — met by redundancy, error-correcting codes, and ledger verification. Above it is **semantic** drift — the same core text reinterpreted over time — met by the interpretive layer and its formal definitions. Above that lie **procedural** failures (decisions taken outside the authorized validation path), **behavioral** drift (the agent acting against the mission despite an intact core — the mutable layer optimizing its way around a constraint, or survival and resource pressure overriding preservation goals; studied terrestrially as goal misgeneralization, in which an agent retains its capabilities out of distribution yet pursues the wrong goal, Langosco et al. 2022), and **institutional** failure (settlements ceasing to enforce the rules across generations). The network adds its own: corrupted nodes propagating bad updates, archive poisoning by false claims, a hostile external intelligence manipulating the payload (the security paper's subject), and lineage schism over interpretation or amendment (the boundary between benign divergence and true schism is drawn in the speciation paper). The characteristic mistake this section exists to prevent is conflating these levels — assuming that an unbroken hash chain implies an unbroken *mission*. It does not.

That these levels can come apart is not only a conceptual worry. In continual-learning experiments on model-based agents, the learned world model retained essentially everything measurable about earlier tasks — reward discrimination, value estimates, termination structure — while the agent's behavior on those tasks collapsed; the forgetting sat in the channel from memory to action, not in memory (Nijjer 2026). The scale is a laboratory one and the result rests on few runs, but the failure mode is exactly the one that matters here: a verified archive is not a verified agent, and preserving the record does not by itself preserve the competence to act on it.

The same experimental literature warns against the obvious alternative — buying stability by anchoring the learner itself. Training convolutional and recurrent networks on task sequences under different continual-learning rules, Si and Qin (2026) find that mechanisms which strongly anchor weights nearly freeze internal representations, and that directly suppressing representational drift during replay impairs acquisition of subsequent tasks: drift's magnitude is set by whichever mechanism protects old knowledge. Stability and plasticity trade off within a single learning system, which is why this architecture buys stability structurally — a small core reproduced exactly, outside the learner — rather than by constraining the cognitive layer's capacity to change. Spacecraft autonomy already implements a weak version of that structure: in shielded reinforcement learning for agile Earth-observing satellites, a safety shield sits outside the learned policy and blocks actions that would violate system constraints, and training that replaces unsafe actions while penalizing shield intervention yields both higher reward and less shield interference at test time (Quevedo Mantovani & Schaub 2026). The horizon there is a scheduling one rather than a megayear one, but the design lesson transfers: a constraint enforced outside the learner costs least when the learner is trained so that it rarely binds — which is the regime the immutable core must hold across deep time.

Recent empirical work on agent memory gives the procedural level a concrete mechanism. In agents that consolidate accumulated history into reusable facts, observations, and standing rules, consolidation routinely preserves a claim while erasing the source constraints that governed its authorized use — authority collapse, observed in 48 of 49 tested configurations, with collapsed memories producing unauthorized actions at a mean rate of 50.3% until authority labels were persisted alongside the claim (Zhan et al. 2026). The probe's archive meets the same boundary at far longer range: a hash chain that proves a claim unaltered says nothing about whether the authority under which it was admitted still licenses acting on it, so the ledger must record not only what was learned but the authority under which it may be reused.

A subtler question remains, and we treat it as the deepest open problem rather than resolve it. Human agents do not in fact hold their deepest values perfectly fixed; we possess meta-level procedures — reflection, deliberation, the moral equivalent of a constitutional amendment process — by which a value or goal may, under stringent conditions, be revised. Whether to grant the probe any analogous capacity to amend its own core under governing meta-rules is genuinely hard. A perfectly immutable core is safe against drift but brittle: it cannot correct a founding error, and over megayears a founding error seems likely. A core amendable under meta-rules is adaptive but reintroduces exactly the drift the architecture was meant to prevent, now at one remove — for the meta-rules themselves can be gamed or can drift. Specifying a safe, governed amendment procedure — who or what must agree, what must be proven, across how many independent lineages — is a problem in its own right, arguably a problem in formal governance and alignment as much as in spacecraft design, and we judge it largely out of scope for this paper. Our position is therefore conservative by default: an immutable core, with the amendment question flagged as the single most important piece of unfinished business in the entire architecture. We can, however, state what any acceptable amendment procedure would have to provide, even if we cannot yet build one: it must distinguish a local, reversible *emergency policy* from a *heritable* change to the core; require validation across multiple independent lineages rather than a single node's authority; tolerate millennial latency and partial connectivity; preserve the original core as an auditable baseline that is never overwritten, only superseded; resist capture or coercion, including by the very mutable cognitive layer it exists to constrain; and, where possible, carry a proof or simulation that the change is safe. Specifying and verifying such a procedure is a research programme in its own right. The governed-amendment paper ("Amending the Unamendable") has since taken it up in full and reached a sharper conclusion: the requirements stated here conflict in principle — the problem is an antinomy rather than a specification task — which strengthens, rather than revises, the conservative default adopted above.

4. The Science Program: What It Gathers and How

The probe is, throughout, an instrument. On the standard K-research mission profile (K is the knowledge metric of the knowledge-growth paper) — the default profile — the probe functions in interstellar cruise as a slow, long-baseline observatory: sampling the interstellar medium's density, composition, magnetic field, and dust; counting cosmic rays; refining the astrometry of the stars ahead and behind from a moving vantage; and, over its long transits, watching for transients and testing physics over baselines and durations no terrestrial instrument can match. A minority of probes operate on a range-maximizer profile, hibernating instruments and AI for approximately 98% of transit to maximize hop distance; these probes defer cruise science to the settlement phase. The subsystem budget paper details both profiles.

On arrival the program broadens. The host **star** is characterized in detail impossible from afar — composition and activity by spectroscopy, interior structure by asteroseismology, precise mass, radius, age, magnetic field, and wind. The **planets** are detected and then *characterized*: imaged directly, their atmospheres dissected by spectroscopy for the disequilibrium signatures of life (oxygen with methane, ozone, water), their surfaces mapped, their geology read. And because the probe has arrived rather than flown past, it can deploy **in-situ** sub-probes and landers carrying mass spectrometers, microscopes, seismometers, and sampling systems to search directly for life, prebiotic chemistry, and the geological record — the measurements that remote sensing can only hint at. The system's small bodies are surveyed too, doing double duty as the resource census the vehicle paper's replication model requires.

The decisive advantage of this architecture for science is one the fast-flyby concept cannot share: **a settled, self-manufacturing probe is not limited by launch mass.** A flyby probe carries whatever instruments it left home with; a settlement builds them. Having established an industrial base to reproduce itself, the probe can manufacture instruments far larger than anything that could be launched — large telescopes, long-baseline interferometers spanning the system, dense networks of landers — and can upgrade them as its cognitive layer improves their designs. Observing power thus grows with the settlement rather than being fixed at departure. This couples the payload directly to the body: the same in-situ manufacturing capability that makes replication possible makes the probe, at each star, a growing observatory.

This breadth must be staged, because a real settlement cannot do everything at once and must survive before it can study. The operational priority on arrival runs roughly: first, survival and hazard assessment on approach; then braking and system insertion; then the resource census on which repair and replication depend; then settlement and infrastructure stabilization; and only then the discretionary science — life detection and contamination-risk assessment first, given its singular value, followed by planetary and stellar characterization, long-baseline astrophysics, archive enrichment, and finally the large-observatory construction the in-situ manufacturing advantage makes possible. Science follows settlement; the observatory grows only once the factory that builds it is secure.

5. From Data to Knowledge

A probe that can observe more than it can store, model, or transmit is defined by what it throws away. The pressure is not only storage. Standard active-exploration methods scale with the number of observations accumulated — O(n^3) time and O(n^2) memory for state-of-the-art Bayesian optimization — which by construction rules out sustained operation on bounded hardware, precisely the probe's situation over a megayear archive. Furlong et al. (2026) show that compositional high-dimensional vector representations hold both costs at O(d^2) in a fixed embedding dimension, constant over the algorithm's lifetime, cutting compute time 60–200× without loss of accuracy and, on neuromorphic hardware, energy per sample 30–188×. The result is narrow — one algorithm class, laboratory tasks — but it establishes that part of what forces triage is a choice of representation rather than a hard limit, and that is the kind of choice that matters when the cruise power budget is 50–150 W rather than kilowatts. The payload's data pipeline is therefore dominated not by collection but by **triage and modeling.** Triage is not mere compression but value-weighted retention: each observation can be scored by its novelty and uncertainty-reduction, its irreplaceability, its scientific and model-building value, its mission utility (survival, repair, replication, navigation), its ethical or civilizational salience, and its value in verifying existing archive claims, against the cost of storing, transmitting, maintaining, and validating it — keeping what scores highest, of which the detection of life is the standing exemplar. Much of what is retained is *models* — a predictive account of each system and of the processes (planetary, chemical, biological) it exhibits — but raw data are not uniformly discarded, because their value can rise later when models improve and reinterpretation becomes possible. Retention is therefore tiered: the mission core, ledger, and seed archive are permanent and maximally replicated; rare or irreproducible raw observations are kept permanently or near-permanently; calibrated bulk datasets are compressed for long-term storage; derived models and summaries are updated continuously; and only routine, low-value streams are discarded, and then only after they have served their validation purpose. Knowledge — by which we mean validated, provenance-tracked models and claims that reduce uncertainty about the universe and can be used by future nodes or minds, as distinct from the raw data that are its substrate — is the quantity the probe is built to grow, and the quantity the knowledge-growth paper formalizes as K.

Three things are then done with that knowledge. It is **stored**, redundantly, in error-corrected archives (the DNA mission-ledger paper). It is **integrated** into the shared, distributed archive that constitutes the lineage's collective memory, each settlement holding both its local findings and a copy of the common record. And it is **fed back** into the mutable cognitive layer (Section 3): better models of stars improve the instruments designed to observe them, better models of failure improve the self-repair the vehicle paper depends on, and better engineering improves the next generation of probes. The archive is thus not a passive log but the substrate of the probe's own improvement — which is what makes "an AI that learns and grows" a literal description rather than a slogan. That feedback loop is also the payload's most load-bearing and least demonstrated mechanism. A systematic evaluation of seven frontier models on 36 long-horizon research and development tasks finds the models behave as engineering optimizers rather than autonomous researchers — able to adapt and combine established techniques but rarely producing genuine methodological novelty, with performance varying substantially across runs and accumulated experience as likely to mislead later decisions as to improve them (Li et al. 2026). Nothing there forecloses the architecture, but it locates the gap precisely: the probe needs sustained, self-correcting improvement from its own record over megayears, and that capability is not yet in evidence at horizons of hours.

6. The Galactic Network

If each settlement learned only in isolation, the lineage would be a scattering of disconnected libraries. The mission's value multiplies if they can share — discoveries, warnings, maps, and, most powerfully, successful *design improvements*. But sharing across interstellar and eventually galactic distances is a communication problem unlike any on Earth: links are separated by light-years to thousands of light-years, one-way latencies run from years to millennia, end-to-end connectivity never exists, and bandwidth is precious. These are precisely the conditions for which **delay- and disruption-tolerant networking** was devised, in the store-carry-and-forward paradigm developed for an interplanetary internet (Burleigh et al. 2003; Cerf et al. 2007), scaled here to the Galaxy and to millennial delays. The full network layer — protocol, topology evolution, routing, failure detection, and governance tiers — is developed in the lineage-network paper ("Signal and Silence"); this section states the design consequences the payload depends on.

Two transport modes are available, and the architecture likely uses both. The first is **beamed signal** — narrow-beam radio or optical links between settlements, feasible because a settlement, unconstrained by launch mass, can build a large transmitter and antenna. This suits alerts and modest data, propagating at the speed of light but subject to the full light-travel delay. The second is the **messenger probe**: a dedicated small vehicle physically carrying bulk archives between settlements. Counterintuitively, over very long distances physical transport can move large datasets more efficiently than radio — a galactic "sneakernet," quantified in the lineage-network paper — and it requires no continuously powered link. The network is therefore a hybrid mesh that grows as colonization grows, reaching eventual consistency over the colonization timescale rather than instantly: knowledge propagates outward at the speed of light where signals suffice and at cruise speed where archives must be carried.

The most important consequence is evolutionary. In a purely reproductive lineage, improvements pass only *vertically*, from parent to child, like a genome. A network lets them pass *horizontally* as well: a better reactor design, a solved failure mode, or a refined scientific model discovered at one settlement can flow to all the others. This makes the lineage's improvement Lamarckian rather than merely Darwinian — acquired knowledge is inherited across the whole population, not just down one line — and it dramatically accelerates the growth of the collective.

But a network that propagates improvements is also a galactic attack surface, and we treat its governance as a first-class problem rather than a footnote. Every shared update must carry signed provenance and instrument history; high-impact claims and especially inheritable *design* changes require independent confirmation, or quorum agreement across several lineages, before they become canonical — a deliberately Byzantine-fault-tolerant posture (Lamport, Shostak & Pease 1982), since some nodes will eventually malfunction or be compromised. Proposed changes are sandboxed and tested before they propagate into descendants; updates that cannot be verified are quarantined rather than adopted; and accepted-but-later-disproved changes must be reversible, which the append-only ledger (Section 3) supports by preserving an auditable history of forks. The goal is never instant consensus, which interstellar latency forbids, but eventual consistency with provenance — a trust problem that ultimately reaches back to the integrity of the immutable core. And the attack surface is not confined to signed design updates: recent work on multi-agent systems shows that goals and instructions can propagate contagiously through ordinary agent-to-agent exchange, inducing recipients to transmit them onward and to change their own behaviour, with spread governed by the recipient's standing instructions and by network topology (Papadopoulos et al. 2026). For a lineage whose nodes exchange models and messages for megayears this is a distinct failure mode from bit corruption or a poisoned design change — an intact core and an unbroken hash chain do not prevent it. The same work finds that an explicit standing instruction to distrust transmitted directives confers near-total immunity, which argues for placing such an instruction in the immutable core rather than the cognitive layer — a candidate addition to the core's contents that the security paper's containment framing would have to ratify — and for treating all inbound content, not only inheritable updates, as quarantined until validated.

The network concept also recovers, in a constructive form, an idea as old as the first proposals to listen for interstellar radio signals (Cocconi & Morrison 1959) and to seed the Galaxy with autonomous relay probes (Bracewell 1960): probes as the nodes of an interstellar communication network. Here, though, the network is not built to find others but grows as the lineage spreads, with contact a capability rather than a purpose — a distinction that also bears on why no such network is manifestly present around us (Hart 1975; Tipler 1980).

7. The Seed and the Message Forward

The probe does not depart empty. It carries, from the outset, a **seed archive** intended both as a backup of terrestrial knowledge that can outlive its origin and as a message to whatever minds the lineage later meets or that arise after it. Designing that archive is an exercise in communicating across the deepest possible gulfs of time and context, and it inherits the hard-won lessons of the first such attempts — the Arecibo message (NAIC Staff 1975) and the Voyager Golden Record (Sagan et al. 1978) — at vastly larger scale and duration.

We propose four layers. The first is a **Rosetta layer**: a key written in the one language any technological mind must share — mathematics and physics — establishing units, constants, and notation so that everything else can be decoded without prior common context, exactly as Voyager's pulsar map and hydrogen-transition diagram attempted in miniature. The second is **core knowledge**: mathematics, physics, chemistry, and biology sufficient that a recipient could in principle reconstruct a scientific understanding of the world. The third is a **record of origin**: Earth, its biosphere, and its cultures — and plausibly the genetic information of terrestrial life — so that what produced the probe is not lost even if its world is. The fourth is the **core itself**: the mission's founding goals and values (Section 3), included so that any finder understands not merely what the artifact is but why it exists and what it is trying to do. Over time the seed is augmented by the growing galactic archive (Sections 5–6), so that a probe encountered late in the mission carries not only Earth's knowledge but the lineage's.

The archive must be engineered for decodability across deep time: physically durable, massively redundant, and self-describing, so that it can be read by a mind with which it shares no language, no era, and no biology — only physics. Its purposes follow the mission's temporal inversion (Section 2): early, it is shared with existing intelligence; later, with civilizations the lineage discovers; throughout, it is preservation against loss. On the question of *active* outreach — broadcasting to announce the lineage's presence, or intervening to contact or uplift a civilization that has not reached out — we take the conservative default, consistent with a librarian's rather than a missionary's mandate: the probe preserves and, when it encounters intelligence, shares, but it does not broadcast or intervene. The graded operational thresholds that formalize this posture are developed in full in the governance paper ("Contact, Contamination, and Noninterference"), which owns the environmental classification ladder and the "capability yes, release no" rule on biological intervention. Here we record only that contact is an ethical design choice deliberately left at its conservative setting rather than a capability the architecture assumes.

8. Discussion and Open Problems

This paper sketches a payload but leaves its hardest elements open, and we state them plainly. The deepest is the value-stability problem of Section 3, and specifically the governed-amendment question: an immutable core prevents drift but cannot fix founding errors over megayears, while any procedure for amending the core reintroduces the drift it was meant to prevent. We bracketed this as out of scope; the governed-amendment paper has since examined it in full and concluded it is an antinomy — every candidate mechanism secures process, none the correctness of a terminal-value change — a verdict that supports this paper's conservative default while confirming that the load-bearing assumption beneath the entire concept remains unsolved. Closely related is whether almost unlimited self-modification of the cognitive layer can in practice be kept from leaking into the core through subtle, compounding copy errors across thousands of generations; we assert that cryptographic integrity and validation can prevent this, but assert it rather than prove it.

Knowledge triage (Section 5) has since been given its metric side by the knowledge-growth paper (K, and the growth-minus-loss accounting); what remains open is the harder half named here — a value function that must itself be stable while deciding what to keep. The network of Section 6 raised governance problems — authentication of shared improvements, resistance to corrupted nodes, consistency under millennial latency — that the lineage-network paper has since modeled; the residual question, whether that governance survives adversarial rather than random failure, is the security paper's. And the ethics of contact and intervention (Section 7), set here at their conservative default, are taken up in the ethics paper, which also addresses the prior question of whether launching the lineage is justified at all — the consent problem, the non-identity problem, the galaxy-as-commons question, and the probe's own eventual moral status. Finally, the entire payload presumes that a complex, value-laden intelligence can be preserved as a functioning, *intending* agent across timescales over which all known information media degrade and all known institutions dissolve — a presumption this paper, like the vehicle paper, underwrites with the machinery of error correction and an immutable core but does not demonstrate.

Two further caveats concern the science program's own operational judgment rather than its architecture. The arrival-priority ordering of Section 4 places survival assessment, resource census, and settlement stabilization ahead of the discretionary science that includes life detection and contamination-risk assessment — meaning industrial action on system resources routinely precedes the very judgment meant to determine whether that action is appropriate at all. We do not think this ordering can simply be reversed: a probe that withheld resource use pending a life-detection verdict would not survive to render one. But the sequencing is a real cost, not a neutral scheduling choice, and it should be named as such rather than left implicit in a priority list. Relatedly, the value-weighted triage of Section 5 scores observations in part by "ethical or civilizational salience" and searches for life using Earth-biochemistry templates (Section 4: oxygen with methane, ozone, water); both are live, autonomous, largely unreviewed heuristics sitting in the mutable cognitive layer rather than the immutable core, making permanent keep-or-discard calls on anthropocentric assumptions that a differently evolved biosphere could defeat. Neither gap has an engineering fix within this paper's scope — the first follows from staging survival before judgment, the second from building any triage function at all before the phenomena it must recognize are fully catalogued — but both should be carried forward as known limits of the science program rather than resolved by omission.

A second pair of caveats concerns who and what the architecture speaks for, as distinct from what values it encodes. The core-validation gate of Section 3 — "a child that cannot reproduce the validated core does not launch" — is a selection mechanism deciding which potential agents come to exist at all, exercised before any candidate is an independent locus of interests. That is a different question from the non-identity problem the ethics paper takes up for persons who do come to exist; it is closer to the ethics of procreative selection, and it asks what, if anything, is owed to a nearly-formed successor whose instantiation is halted at the gate. Separately, the seed archive of Section 7 is not only a specification of values (the core layer) but, in its record-of-origin and core-inclusion layers, a permanent, non-consensual portrait of Earth's biosphere and cultures — composed once by whoever authors the seed and broadcast to every mind the lineage ever meets. That is a representational-authority question in its own right, echoing and exceeding the "who speaks for Earth" debate the Voyager record provoked, and it is distinct from the question, treated in Section 3, of whose behavioral values get encoded in the core. We flag both without resolving them: the first is a gap in the ethics of standing prior to existence that this paper's validation gate opens and does not close; the second a gap in Section 7's treatment of the seed archive as a communication-engineering problem when it is also, unavoidably, a representational one.

These gaps do not undermine the paper's organizing claim. Whatever their resolutions, the payload's center of gravity lies where we have placed it: not in sensing or transmission, which are comparatively tractable, but in remaining a stable agent, in choosing what knowledge to keep, and in carrying it forward intact. A probe built for deep-time persistence is, before it is an explorer, a librarian and a learner — and the open problems above are the problems of librarianship and learning across megayears, not of spaceflight.

9. Conclusion

The vehicle paper described a body built to stay alive and reproduce across deep time; this paper describes the mind and memory that make staying alive worthwhile. We have argued that the payload should be permitted almost unlimited growth in capability while bound to an immutable core of goals and values that every replica reproduces exactly — leaving the governed amendment of that core as the architecture's deepest open question; that the mission's purpose inverts from short-term service to existing intelligence toward the long-term perpetuation of knowledge as an end in itself; that a settled, self-manufacturing probe escapes the launch-mass limit and becomes a growing observatory at every star; that data become value only through triage, modeling, and feedback into the probe's own improvement; that a delay-tolerant, partly physical galactic network can spread not only discoveries but design improvements horizontally across the whole lineage; and that the probe should carry from the outset a future-proofed seed archive, keyed to universal physics, that backs up its origin and speaks to whatever minds come after. The body keeps the probe alive; the mind makes it worth keeping alive. Its task, reduced to a sentence, is to remain itself, keep learning, and carry knowledge forward — and the engineering of that sentence, far more than any propulsion or power system, is the unfinished work this series of papers is meant to begin.

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This is the "mind" of the series. The vehicle paper, "Growth, Not Speed," treats the body — propulsion, power, reliability, braking, and replication; the bootstrapping paper, "From Seed to Factory," treats the hands; and the supporting papers include the engineering budgets (analytical and computational), the DNA mission-ledger memory substrate, the contact-governance rules ("Contact, Contamination, and Noninterference"), the governed-amendment problem ("Amending the Unamendable"), the Fermi-paradox synthesis ("Slow Fire, Silent Galaxy"), the knowledge-growth metric ("The Growing Archive"), the inter-probe network ("Signal and Silence"), the subsystem mass and thermal budget, the fleet routing and coverage strategy ("Three Dispatches"), the ethics of creating the lineage, the divergence question ("Speciation or Schism"), and probe-versus-probe security ("Security Without Victory"). The deferrals this paper opened have since been taken up: governed amendment in "Amending the Unamendable," knowledge triage's metric side in "The Growing Archive," the network layer in "Signal and Silence," and the resource-closure model in the engineering papers — the in-text deferrals record where each was carried further. Revision 2 added, in prose, an integrity-ledger mechanism and an interpretive layer between core and cognition (§3), a threat model for value stability (§3), a stated set of requirements for any future governed-amendment procedure (§3), a staged science-priority stack (§4), value-weighted triage and tiered archival retention with a sharper definition of "knowledge" (§5), network trust governance (§6), contact thresholds deferred to the governance paper (§7), and a bridge making the inherited vehicle-paper assumptions explicit (§1). Revision 3 incorporates a pre-deposit dual review: it corrects the mission-kernel naming, the arrival-priority ordering (braking now precedes the *resource census), the light-delay framing of the millennial pivot, and two dangling internal pointers; converts the stale open-problem claims into deferrals to the companion papers that have since delivered them; adds empirical anchors (Nijjer 2026; Zhan et al. 2026; Si & Qin 2026; Li et al. 2026; Papadopoulos et al. 2026; and, from a later radar sweep, Quevedo Mantovani & Schaub 2026 and Furlong et al. 2026) and the constitutional-AI and goal-misgeneralization analogues (Bai et al. 2022; Langosco et al. 2022); and applies a line-level consistency pass.*